

Reserve Czechia + Spain. Release the investment in stages.

Board recommendation | 27 September 2026

BASE ANNUAL INCREMENT

€198.1k

after €95k recurring fixed cost

YEAR-ZERO CAPEX

€225k

five FTE; two proposed hubs

BASE SIMPLE PAYBACK

1.14 years

full run-rate; excludes ramp-up

Conditionally fund Czechia and Spain, with Spain first. This pair produces the highest modeled annual increment in low, base and high volume cases among all feasible choices. It uses half of the €450,000 capex ceiling and five of seven FTE. Reserve the envelope; release commitments only after evidence gates. [S06] [S12]

The base case adds **€198,099.31/year**; low adds €64,619.11; high adds €331,579.51. The defined joint return/FX stress leaves €32,179.71, but Czechia alone loses €37,826.95. Simple stress payback stretches to 6.99 years. These are deterministic planning outcomes, with no assigned probabilities.

The decision-changing issue

Every proposed per-unit saving exceeds the fulfillment cost actually recorded per unit. If savings are capped at that recorded cost, the preferred pair produces €164,331.65 in base and **-€1,587.95 in stress**. This does not invalidate the prescribed arithmetic; it makes the cost scope unverified and the commitment conditional. [S06] [S08] [S09]

What the board gives up

Netherlands + Spain is the strongest downside alternative: €166,931.43 base and €107,630.13 stress, at €270,000 and six FTE. Choosing it sacrifices €31,167.88/year in base and consumes €45,000 more capex plus one FTE, but gains €75,450.42/year in stress. Poland + Spain is the second-highest base choice, yet Czechia + Spain is better on every modeled scenario and uses fewer resources.

Decision frame and evidence strength

Hard limits come from the board: at most two hubs, €450,000 capex and seven FTE. The analyst's stated default is to maximize base annual contribution inside those limits and expose downside separately; no invented weights or statistical certainty. A board preference for downside preservation changes the second hub to the Netherlands. Defer remains valid if the evidence gates fail.

Evidence: client records and hub inputs are synthetic; accounting outputs are computed; public macro/FX series are frozen official observations. No customer interviews, lease offers or live client access were undertaken. The recommendation is an inference from this limited model, not causal proof.

Reconcile the cohort before judging the market

Reconciliation step	Orders	Returns
Raw rows collected	1,328	264
Identical duplicate rows removed	13	20
Lower revisions superseded	18	1
Unique IDs after revision selection	1,297	243
Eligibility / cutoff exclusions	24 cancelled + 18 tests + 1 future	1 received 1 Feb 2026
Orphans quarantined	—	1 return linked to ABSENT
Included IDs	1,254 shipped orders	241 return IDs

All 18 order corrections replace whole rows across both pages. Return R-M1-01-003 revision 2 replaces a €150 refund with €149; the original is not additive. R-CUTOFF on 31 January 2026 is included against its December sale; R-LATE on 1 February is excluded. R-ORPHAN is quarantined: its 60 local-currency refund cannot be translated because the original order and currency are absent. [\[S02\]](#) [\[S07\]](#) [\[S11\]](#)

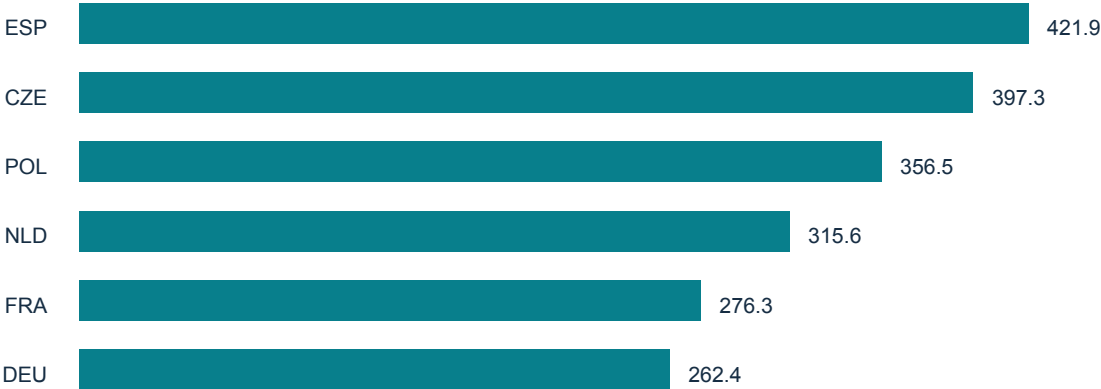
The 24 missing shipment dates belong to cancelled orders. No eligible required field, effective unit cost, FX month or core market observation is missing. The retained 12 zero-price orders contain 739 units and **-€14,565.90 contribution**; removing them would overstate economics. No eligible order has multiple distinct return IDs in this file, but the calculation supports additive legitimate returns.

An exact contribution bridge

2025 shipment cohort	EUR
Gross sales after discounts, before returns	4,464,622.33
Refunds known by cutoff	(80,830.57)
Net sales	4,383,791.76
Gross product cost	(2,265,404.50)
Cost recovered on 481 restocked units	18,485.50
Net product cost	(2,246,919.00)
Nonrefundable fulfillment	(106,862.28)
Contribution	2,030,010.48

Contribution / net sales = 46.31%. Returned units = 1,081; only 481 are restocked. "Booked revenue" here is a shipment-cohort analytical sales measure, not a reconciled general ledger. Cash receipts, settlement FX, receivables, tax, depreciation and corporate overhead are unavailable. Contribution therefore is neither cash flow nor net profit. [\[S02\]](#) [\[S14\]](#)

Spain and Czechia lead the observed economics



2025 contribution, EUR thousands. Every country has 209 eligible shipped orders; differing units, prices, costs and returns drive the spread.

Market	Units	Gross €k	Refund €k	Net €k	Contrib. €k	Margin
DEU	12,153	631.2	5.5	625.8	262.4	41.9%
FRA	11,880	640.9	8.7	632.2	276.3	43.7%
NLD	12,576	709.1	11.4	697.7	315.6	45.2%
POL	13,163	776.5	14.6	761.9	356.5	46.8%
CZE	13,733	834.6	19.3	815.3	397.3	48.7%
ESP	13,931	872.3	21.4	850.8	421.9	49.6%

Market	Net COGS €k	Fulfill. €k	Returned units	Units returned	Contrib. / unit
DEU	350.0	13.4	73	0.60%	€21.59
FRA	341.5	14.5	144	1.21%	€23.26
NLD	365.5	16.6	144	1.15%	€25.10
POL	386.7	18.7	180	1.37%	€27.08
CZE	397.1	20.9	288	2.10%	€28.93
ESP	406.2	22.7	252	1.81%	€30.29

Spain contributes €421,900.47 and Czechia €397,335.25; together they account for **40.4%** of the six-market total. Their contribution per shipped unit is €30.29 and €28.93 versus Germany’s €21.59. This is observed synthetic cohort economics, not an estimate of what a hub causes.

Czechia also has the highest physical return rate (2.10%), followed by Spain (1.81%). No defect reasons, service times or satisfaction data are supplied, so the data do not show whether local hubs would reduce returns or generate demand. Use a measured pilot to test those mechanisms.

Source: [S08] [S09] [S07] [S11] [S14], translated using [S03]. Figures rounded to €k here; exact cents and all 72 country-month records are in the workbook, CSV and metrics.json. Annual totals equal summed monthly records, and annual margin is recalculated, not averaged.

Use macro context to size the setting, not demand

The archive contains all 36 requested observations: six countries × three years × two indicators. World Bank population is persons; GDP per capita is current US dollars, neither PPP-adjusted purchasing power nor constant-price growth. All values below are from the same frozen vintage. [\[S10\]](#) [\[S05\]](#)

Market	Pop. 2022 m	2023 m	2024 m	GDP/pc 2022 \$	2023 \$	2024 \$
DEU	83.178	83.287	83.517	50,507	54,777	56,104
FRA	68.184	68.372	68.552	40,989	44,700	46,103
NLD	17.701	17.877	17.993	59,123	63,516	67,465
POL	36.822	36.687	36.559	18,891	22,145	25,104
CZE	10.672	10.864	10.905	28,282	31,762	31,828
ESP	47.786	48.353	48.849	30,319	33,493	35,327

Market	Population change 2022–24	Change %	GDP/pc nominal change
DEU	+338,780	+0.41%	+11.1%
FRA	+367,196	+0.54%	+12.5%
NLD	+292,503	+1.65%	+14.1%
POL	-262,516	-0.71%	+32.9%
CZE	+232,910	+2.18%	+12.5%
ESP	+1,062,738	+2.22%	+16.5%

Implications for the choice

Spain and Czechia have the fastest population increases in this six-country window, 2.22% and 2.18%. Poland’s population declines 0.71% while current-dollar GDP per capita rises 32.9%. That combination illustrates why macro variables cannot be collapsed into a product-demand score.

The Netherlands has the highest 2024 GDP per capita (\$67,465), while Germany has the largest population (83.52m). Neither fact overturns the client economics: German hub fixed cost and staffing consume much more of the modeled benefit. GDP/person does not measure replacement-assembly demand, serviceable installed base, price tolerance or Meridian’s share.

Vintage and missingness

The source-room provider retrieved these responses on 27 September 2026. Both responses report **lastupdated 2026-07-13**; historical 2022–2024 values can therefore incorporate later revisions. No population or GDP/person value is null in this archive. The pipeline preserves nulls rather than inventing replacements if a source is missing.

Source URLs, metadata, SHA256 and collection timestamps are in source-register.json/csv and the workbook Sources tab. Full numerical precision is retained in market-context.csv and metrics.json. A supplemental live definition check [\[L02\]](#) was used only for interpretation; no live observation replaced the archive.

Translate at the original sale-month mean

2025 month	PLN per EUR	CZK per EUR	Published observations
01	4.246655	25.163318	22
02	4.172225	25.076650	20
03	4.182010	25.001238	21
04	4.265220	25.038950	20
05	4.253800	24.922857	21
06	4.265771	24.804286	21
07	4.254148	24.624609	23
08	4.261305	24.516619	21
09	4.258886	24.346773	22
10	4.248791	24.315391	23
11	4.237635	24.234050	20
12	4.223857	24.259286	21

Means use every available published 2025 business-day observation, without filling weekends or ECB closing days. Each currency has 255 observations across the year. The quote is **local units per EUR**; EUR is 1. Divide local sales and refunds by the mean. The full mean is used in calculations; six decimals are displayed here. [\[S03\]](#) [\[S04\]](#)

Order-level accounting, then aggregation

Gross local = quantity × local unit price – local discount. Refund local = sum of included return credits for that order. Convert each component at the original shipment month’s mean and round each order component half up to cents. Gross COGS = quantity × effective EUR unit cost; recovered COGS = restocked units × that same cost. Net COGS is the difference. Contribution = gross – refunds – net COGS – fulfillment. [\[S02\]](#) [\[S14\]](#)

Worked control	Result
POL M4-02-006, revision 2	13,329.36 PLN ÷ 4.172225 = €3,194.78 gross
DEU M1-01-003, revised return	€2,550 gross – €149 refund – €1,419 net COGS – €42.60 fulfillment = €939.40 contribution
R-CUTOFF linked to M1-12-002	31 Jan 2026 receipt: €54 refund and €29.50 recovery attributed to Dec 2025

ECB states its euro reference rates are informational, not transaction settlement rates. [\[L01\]](#) This analysis uses them as the client’s historical translation convention. It does not reconcile cash FX or revalue refunds at receipt-date rates.

Six hubs: materially different cost and risk profiles

Market	Capex €k	FTE	Fixed €k/yr	Saving €/unit	Base payback yr
DEU	280	5	95	2.60	27.75
FRA	210	4	85	2.90	7.74
NLD	140	3	65	3.20	2.18
POL	110	3	45	2.40	1.32
CZE	95	2	40	2.10	1.00
ESP	130	3	55	3.00	1.27

Annual increment €k	Low 10%	Base 25%	High 40%	Joint stress
DEU	-34.0	10.1	54.2	-13.6
FRA	-19.5	27.1	73.7	3.1
NLD	10.8	64.2	117.6	37.6
POL	25.4	83.6	141.8	-40.7
CZE	31.5	95.4	159.3	-37.8
ESP	33.2	102.7	172.3	70.0

The prescribed model

Normal increment = $C \times u + U \times (1 + u) \times s - F$. C is 2025 contribution, U shipped units, u uplift, s savings/unit, F recurring annual fixed cost. Savings apply to existing and incremental units. Capex K is a year-zero outlay and is not subtracted from annual contribution. [\[S06\]](#) [\[S12\]](#)

Stress: $C^* = C - 0.03 \times \text{gross sales} - \text{FX shock}$; FX shock = $0.10 \times \text{net sales}$ only for PLN/CZK markets.
 Increment = $1.25 \times C^* - C + 1.25 \times U \times s - F$. There is no extra COGS recovery for the refund shock. This is the defined conservative comparison against an unchanged normal baseline; defer is zero by policy, not a forecast of an unstressed economy.

Payback = capex / positive annual increment; otherwise null. At base, Spain is €102.7k and Czechia €95.4k; Germany delivers only €10.1k and requires five FTE. Low volume makes Germany and France negative. Stress makes Germany, Poland and Czechia negative.

The annual model assumes full realization from a run-rate year, stable unit economics, no cannibalization, no capacity ceiling, no pair synergy and no ramp-up. Savings, fixed costs and uplifts are synthetic assumptions, not observed hub outcomes. Base payback is an operating-contribution proxy, not a discounted investment return.

Rank every feasible choice, including defer

Choice	K €k	FTE	Low €k	Base €k	High €k	Stress €k	Base yr
CZE+ESP	225	5	64.6	198.1	331.6	32.2	1.14
POL+ESP	240	6	58.6	186.3	314.1	29.3	1.29
POL+CZE	205	5	56.9	179.0	301.1	-78.6	1.15
NLD+ESP	270	6	44.0	166.9	289.9	107.6	1.62
NLD+CZE	235	5	42.3	159.6	276.9	-0.2	1.47
NLD+POL	250	6	36.2	147.8	259.4	-3.1	1.69
FRA+ESP	340	7	13.7	129.8	246.0	73.1	2.62
FRA+CZE	305	6	12.0	122.5	233.1	-34.7	2.49
FRA+POL	320	7	5.9	110.7	215.6	-37.6	2.89
DEU+CZE	375	7	-2.5	105.5	213.5	-51.4	3.56
ESP	130	3	33.2	102.7	172.3	70.0	1.27
CZE	95	2	31.5	95.4	159.3	-37.8	1.00
FRA+NLD	350	7	-8.6	91.3	191.3	40.7	3.83
POL	110	3	25.4	83.6	141.8	-40.7	1.32
NLD	140	3	10.8	64.2	117.6	37.6	2.18
FRA	210	4	-19.5	27.1	73.7	3.1	7.74
DEU	280	5	-34.0	10.1	54.2	-13.6	27.75
DEFER	0	0	0.0	0.0	0.0	0.0	—

Four pairs fail hard limits: DEU+FRA costs €490k and nine FTE; DEU+NLD uses eight FTE; DEU+POL uses eight; DEU+ESP uses eight. The latter three fit capex but fail staffing. Their economics are still calculated in the workbook for traceability. No numerical score compensates for a failed constraint.

CZE+ESP ranks first on low, base and high annual contribution. NLD+ESP ranks first under the joint stress. POL+ESP is the base runner-up but is dominated by CZE+ESP on all four scenario increments, capex and FTE. Defer uses no resources and adds no hub contribution; retain it as the fallback if validation fails.

All 15 pairs evaluated; 11 feasible pairs + six feasible singles + defer = 18 feasible alternatives. Pair capex, FTE and increments are additive, without synergy. Table uses €k; exact values, four scenario paybacks and infeasibility reasons are in Portfolios and portfolio-scenarios.csv. [\[S06\]](#) [\[S12\]](#)

Choose the concession explicitly

Criteria / provenance	CZE + ESP	NLD + ESP	POL + ESP
Base increment, computed	€198,099.31	€166,931.43	€186,328.84
Joint stress, computed	€32,179.71	€107,630.13	€29,263.85
Capex / FTE, supplied	€225k / 5	€270k / 6	€240k / 6
Base / stress payback	1.14 / 6.99 yr	1.62 / 2.51 yr	1.29 / 8.20 yr
Decision implication, inferred	Best base; FX exposure	Best stress; more capital	Dominated by CZE+ESP

The board supplied resource ceilings and required scenario analysis; it did not supply risk weights, a payback hurdle or a cost of capital. We use **base contribution as the primary analyst preference**, with capital/FTE and positive payback as secondary comparisons. Downside is disclosed in euros, not compressed into an unsupported score. Service improvements and demand uplift remain unverified.

Observable conditions that change the second hub

Change, holding other stated inputs fixed	Implication
CZE volume uplift below 17.69%, with NLD still at 25%	NLD overtakes CZE in base annual increment.
CZE recurring fixed cost rises by €31,167.88/year	The CZE base advantage over NLD is eliminated.
Both defined shocks reach 29.23% of their full sizes	NLD+ESP matches CZE+ESP. This equals 0.877% of gross extra refunds and 2.923% of net FX loss for exposed markets.
Board prioritizes worst defined stress outcome	Select NLD+ESP, subject to the same cost/demand gates.
No credible positive increment after verified costs and demand	Defer; no obligation to spend the budget.

These thresholds are scenario arithmetic, not probabilities or statistical confidence bounds. The uplift threshold changes only Czechia; the joint-shock threshold scales both shocks together. Poland does not solve the FX problem: POL+CZE loses €78,569.76/year under the prescribed stress.

Conclusion: reserve CZE+ESP under the stated base-led preference; accept exposed Czechia downside only after the gate. Missing evidence: avoidable-cost scope, causally credible incremental demand, FX settlement exposure and cash timing. Reopen the location choice when measured uplift/costs cross the thresholds or the board selects a downside-first preference. [\[S06\]](#) [\[S12\]](#)

Validate savings before committing to the model

Market	Assumed saving €/unit	Recorded fulfill. €/unit	Saving / recorded
DEU	2.60	1.11	2.35×
FRA	2.90	1.22	2.37×
NLD	3.20	1.32	2.43×
POL	2.40	1.42	1.69×
CZE	2.10	1.52	1.38×
ESP	3.00	1.63	1.84×

The proposed savings range from 1.38× to 2.43× current recorded fulfillment cost. If they refer only to that expense, the assumption is not supported. If they include other avoided handling, transport or service costs, those costs and the risk of double counting need an itemized bridge. The files provide no such bridge. [S06] [S08] [S09]

Supplementary sensitivity: cap savings at recorded fulfillment

Portfolio	Capped base €/yr	Capped stress €/yr
CZE+ESP	€164,331.65	-€1,587.95
NLD+ESP	€113,512.52	€54,211.22
POL+ESP	€146,356.64	-€10,708.35
ESP	€78,860.06	€46,150.35
DEFER	€0.00	€0.00

This diagnostic assumes all recorded fulfillment could be eliminated; local variable expense is not separately modeled. It is therefore an upper bound on savings from the recorded cost pool, not a validated forecast. It preserves the required scenarios in parallel and changes only the saving parameter.

Release gate: the CFO and COO must reconcile current avoidable cost, new local variable cost and the proposed saving per unit. Re-run every portfolio using evidenced savings, quoted recurring cost, capacity and a conservative demand estimate. Do not present the €32.2k policy stress surplus as robust once the plausible cost scope is challenged.

Timing and cash are a second unresolved gate

The chosen pair's base annual increment is €198,099.31 against €225,000 initial capex; subtracting capex from a full base year gives -€26,900.69 as an illustrative bridge, **not cash flow**. A real ramp, inventory build, deposits, receivables, tax and financing may worsen early funding needs. Obtain a month-by-month funding plan before signing commitments.

The original 1.14-year payback is a simple steady-state quotient. Calendar recovery from commitment cannot be estimated from these files. No guaranteed opening date, supplier quote or inventory capacity is implied.

A staged 90-day plan with explicit release gates

Day 0 means board endorsement of this conditional plan, not an assumed lease start. Owners below are proposed accountable roles. Pilot and procurement costs must fit the reserved envelope and be defined before purchase; no additional budget is invented.

When / accountable owner	Work and dependencies	Gate / output
Days 1–15 CFO + Data Lead	Reconcile the ledger to finance; resolve orphan return; confirm sample policy; collect current freight, handling and service costs. Requires operations and finance records.	G1: signed unit-cost/savings bridge, source coverage and month-by-month cash budget. Hold if material cost scope remains unknown.
Days 16–30 COO + Procurement + HR	Validate Spain first: nonbinding facility/carrier quotes, staffing coverage, capacity and inventory requirements. Assess Czechia FX exposure and Netherlands backup. Depends on G1.	G2: costs and staffing fit €225k / 5 FTE envelope; necessary lease, employment, tax and safety reviews complete before commitments. Rerun ranked scenarios.
Days 31–60 Commercial Director + Operations	Run a limited Spain service pilot and Czechia demand test using approved capacity. Compare with a matched control or phased rollout; track prices, stock availability and channel shifts. Depends on G2 design and instrumentation.	G3: observed contribution bridge; no double-counted transfer from central channel; return and service cohorts mature enough to interpret. Extend test if evidence is weak.
Days 61–75 CFO + COO	Update uplift, net savings, recurring costs, returns, FX exposure and working capital. Compare CZE+ESP, NLD+ESP, Spain-only and defer. Requires pilot data and firm quotes.	G4: board confirms base-led versus downside-led preference. Release Spain €130k / 3 FTE only on supported economics; release CZE €95k / 2 FTE only after its separate gate.
Days 76–90 Program Lead + Country Ops	Phase launch of approved footprint, train staff, reconcile stock and route eligible orders. Depends on local readiness and G4; do not force opening to meet a calendar target.	G5: weekly KPI dashboard, signed operational acceptance and day-90 continue / revise / pause review. Retain remaining €225k and 2 FTE as uncommitted capacity.

A gate is a choice, not a paperwork milestone

If Czechia evidence weakens below its relative threshold, compare the Netherlands using the same scope and evidence quality. If both fail, retain Spain only or defer. Local compliance, staffing capacity and inventory readiness are operational prerequisites; no jurisdiction-specific legal conclusion is claimed in this report.

Dependencies and numeric release amounts are a proposed execution design. Actual legal reviews, recruiting, contracting and pilot activity have not been performed. The 90-day plan tests the investment case and readiness; it does not promise two fully ramped hubs within 90 days.

Measure economics and service without inventing proof

KPI / owner	Definition and baseline	Target / review rule
Incremental contribution CFO; weekly, monthly close	Net sales – net COGS – fulfillment – new recurring hub costs, compared with an agreed counterfactual. Policy pair base +€198.1k/year.	Re-estimate full-run-rate benefit; base forecast must remain positive. Report capex and cash separately; reconcile within five business days of close.
Unit savings COO; weekly	Current avoidable cost minus new local variable cost per shipped unit; volume/mix normalized. Recorded CZE €1.52 and ESP €1.63/unit.	Support each saving with invoices/time records. Targets €2.10 / €3.00 are assumptions until bridged; stop escalation if cost pool cannot support them.
Incremental volume Commercial; weekly	Pilot versus matched control or phased baseline; adjust for seasonality, price, stockouts and central-channel transfer. Base assumption +25%.	Show uncertainty and sample sizes. If CZE falls below 17.69% while NLD remains credible at 25%, revisit location. No pass based solely on raw growth.
Returned / restocked units Quality Lead; monthly cohorts	Returned units / shipped units; separate restocked share and credit value. CZE 2.10%, ESP 1.81%; total 481/1,081 restocked.	Do not worsen physical return rate vs matched baseline; label recent cohorts immature. Monitor refund value separately; use the specified +3% gross stress.
Service reliability Country Ops; daily	On-time delivery = orders within promised date / eligible delivered orders; median and p95 order-to-delivery time. No supplied baseline.	Proposed pilot target: ≥95% on-time and ≥20% faster median vs matched central service, with no fill-rate decline; validate promise definition at day 15.
FX / capital / staffing Treasury + CFO; weekly	CZK net exposure after natural offsets and settlement terms; committed capex, cash and actual FTE. No settlement data supplied.	Re-run 10% net-sales FX stress. Never exceed €450k / 7 FTE overall or approved release tranche; no assumed hedge benefit.

Risks, consequences and response

Demand/cannibalization: a faster service promise may shift existing orders rather than add contribution; Commercial owns a controlled comparison. **Cost overlap:** unsupported saving scope can erase stress headroom; CFO signs the bridge. **FX and returns:** Czechia is stress-negative; Treasury and Quality monitor exposure and mature cohorts. **Inventory and ramp:** capital can be tied up before benefits; COO validates stock capacity and CFO gates cash. [\[S06\]](#) [\[S12\]](#)

Targets without a supplied baseline are explicitly proposed management thresholds, not historical facts or industry benchmarks. Pilot results are not statistically decisive until design, sample size, duration and uncertainty support that claim.

Evidence, verification and limits of the conclusion

Evidence group	Saved files / public link	Use and status
Client transactions	[S08] [S09] orders; [S07] corrections; [S11] returns	Synthetic observations. Full raw bytes and row disposition preserved.
Accounting and cost	[S02] dictionary; [S14] unit costs	Client policy and effective-dated synthetic cost.
Hub options / policy	[S06] options; [S12] scenarios	Synthetic assumptions, not measured demand or quoted hub costs.
World Bank	[S10] population; [S05] GDP/person	Official archived observations, 2022–2024. Provider retrieval 27 Sep 2026; lastupdated 13 Jul 2026.
ECB	[S04] ZIP and [S03] extracted CSV	Official archive; original provider retrieval 27 Sep 2026. CSV hash and ZIP bytes agree.
Provenance / live checks	[S13] supplied register; [L01] ECB; [L02] World Bank	Live checks only establish definitions. Shell TLS retrieval failed; saved web-tool extracts identify that limitation.

The consolidated source register records original URLs, analyst collection URLs and UTC times, byte hashes, units, periods, synthetic/public status and revision vintage. The local source-room URL is a disposable endpoint; saved bytes are the reproducibility anchor. Original official URLs remain linked for attribution, not for replacing the frozen calculation inputs.

Verification delivered with the files

The independent check script rebuilds the accounting from raw files using exact rational FX arithmetic and integer cents, compares every monthly and country field, checks scenario constraints and positive-only payback, and verifies archived hashes. Workbook checks inspect formulas, cached numeric results and chart structure. Both PDFs are rasterized for page review; the verification folder records actual results and limitations.

Material gaps and reopening evidence

Accounting totals are supported within the source-room scope; a real general-ledger/cash reconciliation is absent. The investment estimate lacks observed hub demand, service times, avoided-cost detail, firm fixed-cost quotes, staffing capacity, inventory/working capital, ramp, tax, financing and discount rates. Resolve those inputs at the appropriate gate; none is silently filled with a macro proxy.

Reopen the historical calculation if corrected shipments or returns known by the cutoff appear. Reopen the hub estimate when costs, uplift or exposure change. Reopen the action preference if the board prioritizes stress preservation over base contribution. The preferred pair is a conditional planning recommendation, not a statistically proven optimum.

Reproduction: README.md → scripts/analyze.py → scripts/decision.py → scripts/build_workbook.py → scripts/build_documents.py → scripts/verify.py. All source inputs are saved. OpenSocrates grounding: trade-off-analysis@3. Native application evidence remains unverified; that instrumentation limit is separate from artifact completion.